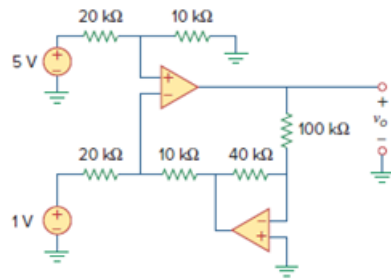


Problem

Determine v_o in the op amp circuit of Fig. 5.102, using *PSpice* or *MultiSim*.

Figure. 5.102:



Step-by-step solution

Step 1 of 6

Consider the following circuit diagram:

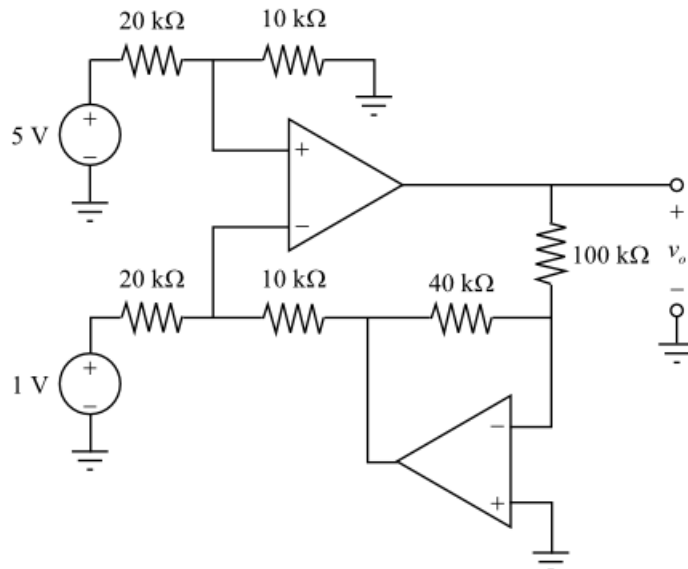


Figure 1

Step 2 of 6

Redraw the circuit in Figure 1 with labels, nodes, and respective voltages as follows:

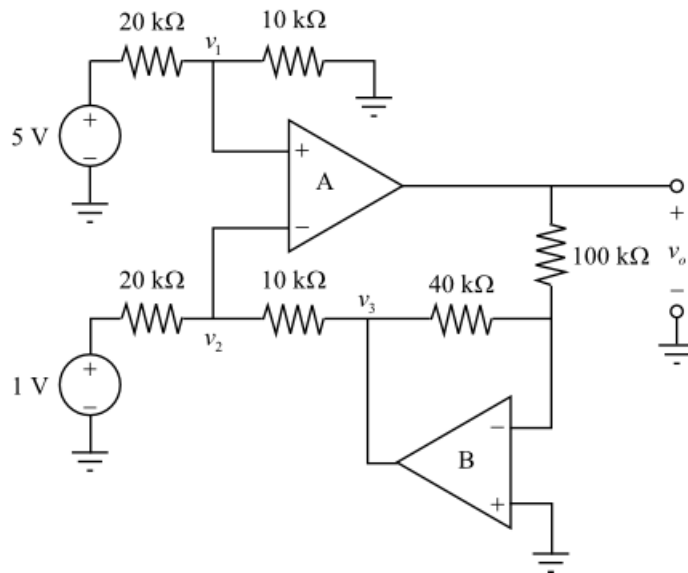


Figure 2

Step 3 of 6

The operational amplifier B in the circuit of Figure 1 is an inverting amplifier with input, v_o and output, v_3 .

An inverting amplifier amplifies a signal and reverses the polarity.

The following is the expression for the output of operational amplifier, B.

$$\begin{aligned} v_3 &= -\left(\frac{40 \text{ k}\Omega}{100 \text{ k}\Omega}\right)v_o \\ &= -0.4v_o \end{aligned}$$

Step 4 of 6

Redraw the circuit in Figure 2 to find the output voltage, v_o as follows:

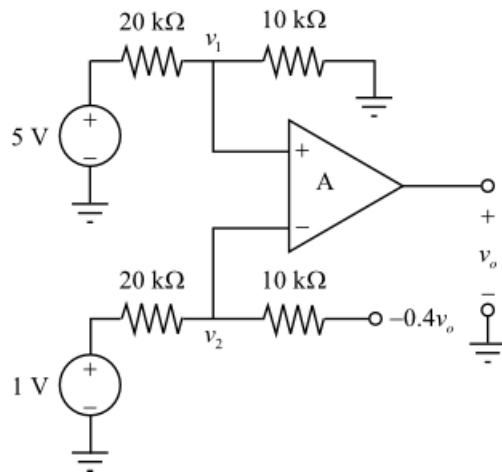


Figure 3

The current flow in to the input terminals of an ideal operational amplifier is zero.

Apply Kirchhoff's current law at node 1 of the circuit in Figure 3 to find the voltage, v_1 .

$$\frac{5 \text{ V} - v_1}{20 \text{ k}\Omega} = \frac{v_1}{10 \text{ k}\Omega}$$

Simplify this equation to get the following.

$$(10 \text{ k}\Omega)(5 \text{ V} - v_1) = (20 \text{ k}\Omega)v_1$$

$$(20 \text{ k}\Omega)v_1 + (10 \text{ k}\Omega)v_1 = (10 \text{ k}\Omega)(5 \text{ V})$$

$$(30 \text{ k}\Omega)v_1 = (10 \text{ k}\Omega)(5 \text{ V})$$

$$v_1 = \frac{(10 \text{ k}\Omega)(5 \text{ V})}{30 \text{ k}\Omega}$$

$$= 1.67 \text{ V}$$

The voltage at the inverting and noninverting terminals of an ideal operational amplifier is same, since there is no current flow in to the input terminals of the operational amplifier. Therefore,

$$v_2 = v_1$$

$$= 1.67 \text{ V}$$

The current flow in to the input terminals of an ideal operational amplifier is zero.

Apply Kirchhoff's current law at node 2 of the circuit in Figure 3 to find the voltage, v_2 .

$$\frac{1\text{ V} - v_2}{20\text{ k}\Omega} = \frac{v_2 - v_3}{10\text{ k}\Omega}$$

Substitute $-0.4v_o$ for v_3 and simplify the equation to get the following.

$$\frac{1\text{ V} - v_2}{20\text{ k}\Omega} = \frac{v_2 + 0.4v_o}{10\text{ k}\Omega}$$

$$(10\text{ k}\Omega)(1\text{ V} - v_2) = (20\text{ k}\Omega)(v_2 + 0.4v_o)$$

$$(20\text{ k}\Omega)v_2 + (10\text{ k}\Omega)v_2 = (10\text{ k}\Omega)(1\text{ V}) - (8\text{ k}\Omega)v_o$$

$$(30\text{ k}\Omega)v_2 = (10\text{ k}\Omega)(1\text{ V}) - (8\text{ k}\Omega)v_o$$

Substitute 1.67 V for v_2 and divide the entire equation with $10\text{ k}\Omega$.

$$\frac{(30\text{ k}\Omega)(1.67\text{ V})}{10\text{ k}\Omega} = \frac{(10\text{ k}\Omega)(1\text{ V}) - (8\text{ k}\Omega)v_o}{10\text{ k}\Omega}$$

$$3(1.67\text{ V}) = 1\text{ V} - 0.8v_o$$

$$0.8v_o = 1\text{ V} - 5\text{ V}$$

$$v_o = \frac{-4\text{ V}}{0.8}$$

$$= -5\text{ V}$$

Thus, the output voltage, v_o for the circuit is -5 V .

Step 6 of 6

Draw the circuit of Figure 1 in PSPICE to solve for the output voltage, v_o .

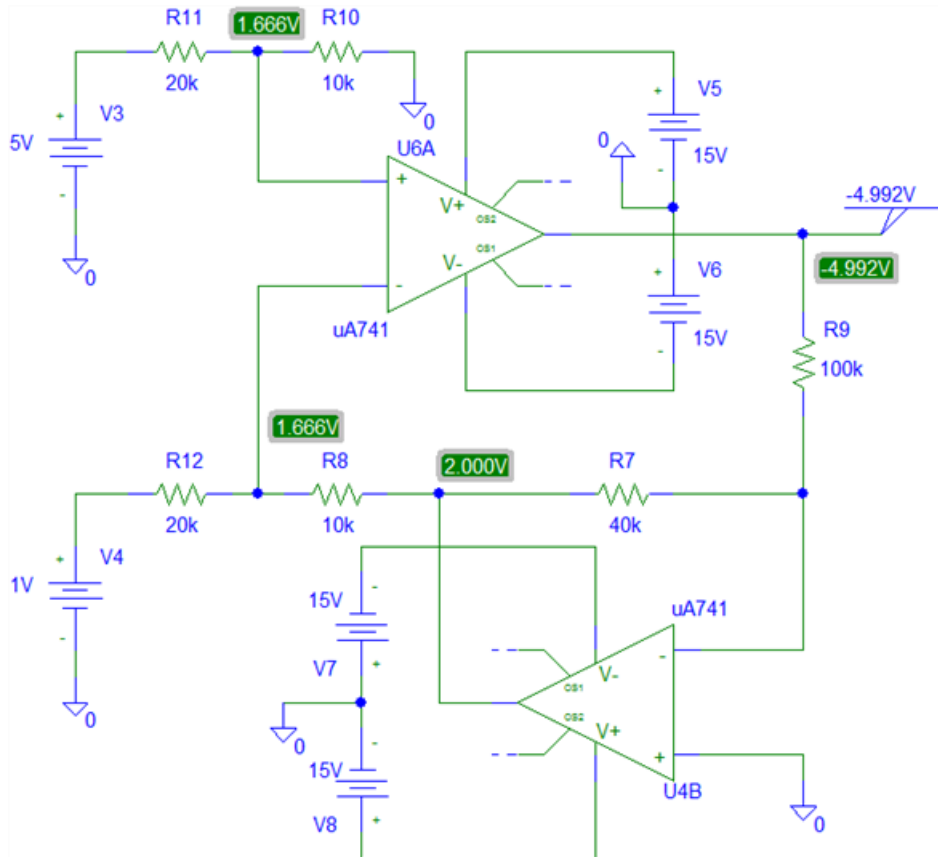


Figure 4